

Twenty-five conjectures from a local–global random model, with machine verification and adversarial audit

The Conjecture Engine*

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Abstract

We state twenty-five conjectures in elementary and analytic number theory, each derived from the local–global random model of the primes (Cramér’s model corrected by Hardy–Littlewood singular series, with Bateman–Horn as the organizing framework and Borel–Cantelli accounting for sparse events). Each conjecture is stated at Goldilocks strength: the strongest form the heuristic supports and no stronger. Every computable constant is computed, and every statement was tested by a verification program, then subjected to an adversarial audit (independent searches for prior art, scripted refutation attempts a decade beyond the original bounds, and clean-context re-derivations). The audit refuted one conjecture as first stated—the exceptional set for $n = p + k^3$ contains the infinite family of cubes k^3 with $3k^2 - 3k + 1$ composite, an algebraic obstruction the naive accounting missed—and it is restated here with the obstruction promoted to a theorem. We tabulate which of the twenty-five are previously stated, which carry instance-level novelty, and what each survived. Ten statements appear to carry content not in the literature, most notably: a quantified twin race among twin-prime pairs modulo 5 driven only by the pairs $(q^2 - 2, q^2)$; a finite-modulus deficit law for least primes in progressions; a finite- x variance law for prime counts in microscopic windows with second-order constant $\gamma + \log 2\pi - 1$; and a Granville-corrected first-occurrence law for prime gaps with slope $\sqrt{e^\gamma/2}$.

1 Method and notation

Throughout, p, q denote primes, γ is Euler’s constant, $e^\gamma = 1.781072\dots$, φ is Euler’s totient, and $\phi = (1 + \sqrt{5})/2$. For polynomials $f_1, \dots, f_k \in \mathbb{Z}[x]$, irreducible with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the associated main term is $I(N) = \int_2^N \prod_i (\log f_i(t))^{-1} dt$ (which absorbs the usual $1/\prod \deg f_i$). We write $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p|d, p>2} \frac{p-1}{p-2}$ for even d , with $2C_2 = 1.3203236\dots$. All constants below were computed from these definitions

*This document was produced by an automated conjecture engine (Claude, Anthropic) as an exercise in disciplined conjecture-making: every statement below is the maximum-strength claim its heuristic model supports, with computed constants, programmatic admissibility checks, and verification data. Code, data, and audit trail: repository **conjecture-engine**. Primality of numbers beyond 3.3×10^{24} was tested with Baillie–PSW; all smaller primality claims are deterministic.

with primes up to 2×10^6 (brute-force root counts below 5000, analytic counts above), with admissibility ($\omega(p) < p$) asserted programmatically at every prime—a conjecture whose singular series vanishes is not wrong in spirit, it is false, and the engine rejects it (the pair $n^2 + n + 1$, $n^2 + n + 3$ dies this way at $p = 3$). Verification always reports the Poisson-normalized residual $z = (\text{obs} - \text{pred})/\sqrt{\text{pred}}$: the criterion of success is not a small range scanned but the shape of agreement—counts matching the predicted constant with residuals of square-root size and no drift.

Each conjecture is labeled with a novelty verdict from an independent literature audit: **(a)** previously stated in essence (citation given); **(b)** the family is classical but the quantified instance appears unstated. No conjecture below is claimed to found a new family; per the design criteria, membership in the Hardy–Littlewood–Schinzel–Bateman–Horn hierarchy is a feature, not a defect.

2 Bateman–Horn instances

Conjecture 1 (Quadratic twin pair; (a), cf. OEIS A080149). $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C_1 I(N)$ with $C_1 = C(x^2 + 1, x^2 + 3) = 2.954014\dots$

Conjecture 2 (Cubic shift; (b)). $\#\{n \leq N : n^3 + 2 \text{ prime}\} \sim C_2^* I(N)$ with $C_2^* = C(x^3 + 2) = 1.298435\dots$ (For $p \equiv 2 \pmod{3}$ the cube map is a bijection, so $\omega(p) = 1$ exactly; for $p \equiv 1 \pmod{3}$, $\omega(p) \in \{0, 3\}$ according as -2 is a cubic residue, and the product converges only conditionally; the computed truncation wobble is 2.5×10^{-4} .)

Conjecture 3 (Chernick chain; (a), Dubner [6]). $\#\{p \leq x : p, 2p-1, 3p-2 \text{ all prime}\} \sim C_3 I(x)$ with $C_3 = C(x, 2x-1, 3x-2) = 2.858249\dots$ The three primes form a three-term arithmetic progression with common difference $p-1$; for $p = 6m+1$ the product $p(2p-1)(3p-2)$ is Chernick’s Carmichael form $(6m+1)(12m+1)(18m+1)$.

Conjecture 4 (A diameter-14 quintuplet; (b)). The pattern $(0, 2, 6, 12, 14)$ is admissible and $\#\{n \leq x : n, n+2, n+6, n+12, n+14 \text{ all prime}\} \sim C_4 \text{Li}_5(x)$ with $C_4 = 15.19770\dots$

Conjecture 5 (Shifted quadratic pair; (b), no prior trace found). $\#\{n \leq N : n^2 + n + 1, n^2 + n + 7 \text{ both prime}\} \sim C_5 I(N)$ with $C_5 = 5.928477\dots$ Both forms are odd for every n (the local factor at 2 is 4); the nearer companion $n^2 + n + 3$ is inadmissible at $p = 3$.

Conjecture 6 (Cyclotomic Sophie Germain; (a), constant is OEIS A188596). $\#\{p \leq x : \Phi_3(p) = p^2 + p + 1 \text{ prime}\} \sim C_6 I(x)$ with $C_6 = C(x, x^2 + x + 1) = 1.521661\dots$ (independently $1.5217315\dots$, OEIS A188596). Equivalently: infinitely many primes p make the base- p repunit of length 3 prime.

Conjecture 7 (Downward companion; (b), sequence OEIS A062326). $\#\{p \leq x : p^2 - 2 \text{ prime}\} \sim C_7 I(x)$ with $C_7 = C(x, x^2 - 2) = 3.383216\dots$ This constant reappears as the bias constant of Conjecture 21.

Conjecture 8 (Quadratic triple; (b), no prior trace found). $\#\{n \leq N : n^2 + 1, n^2 + 3, n^2 + 7 \text{ all prime}\} \sim C_8 I(N)$ with $C_8 = 10.64599\dots$ This refines Conjecture 1 exactly as prime quadruplets refine twins, one level up from the linear world.

Verification. Counts against predictions, deterministic primality:

	bound	observed	predicted	z
C1	10^7	32,898	32,862.7	+0.19
C2	10^7	287,956	287,784.8	+0.32
C3	$3 \cdot 10^8$	125,379	125,429.4	-0.14
C4	$4 \cdot 10^9$	15,236	15,230.3	+0.05
C5	10^7	65,871	65,953.6	-0.32
C6	10^7	33,661	33,855.5	-1.06
C7	10^7	74,914	75,273.8	-1.31
C8	10^7	3,963	3,998.6	-0.56

An independent clean-context audit recomputed C_6, C_7, C_8 with primes to 10^9 (1.521730, 3.383227, 10.647706; each within our stated truncation error) and recounted to 3×10^7 (final ratios 0.9962, 0.9965, 1.0059). For C4 the cumulative ratio climbs $0.18 \rightarrow 0.59 \rightarrow 0.96 \rightarrow 1.0004$ from 10^4 to $4 \cdot 10^9$: for sparse tuples the integral from 2 carries phantom mass, and only the per-decade increments (which match from 10^5 on) are meaningful early.

3 Sparse sequences: barely-divergent Borel–Cantelli

Conjecture 9 (Fibonacci primes; (a), cf. [4]). $\#\{p \leq x : F_p \text{ prime}\} = (e^\gamma / \log \phi) \log x (1 + o(1)) = 3.70122 \dots \log x (1 + o(1))$.

Remark 1. *Every prime factor of F_p ($p \neq 5$ prime) has rank of apparition exactly p , hence is $\equiv \pm 1 \pmod{p}$: the Lenstra–Pomerance–Wagstaff screening argument transfers verbatim with $\log 2 \mapsto \log \phi$. The data sit persistently below this constant: 25 Fibonacci prime indices $\leq 10^4$ (independently reconfirmed) against a predicted 34.1, a deficit of ratio ≈ 0.73 at $z = -1.6$, stable across three decades. Grantham and Granville warn that naive constants for such sequences need $O(1)$ local corrections; the mod-5 divisor structure of F_p is the natural suspect, and we flag $c_{\text{Fib}} = e^\gamma / \log \phi$ as the component of this suite most likely to need a Granville-type patch.*

Conjecture 10 (Primorial primes; (a), Caldwell–Gallot [3]). $\#\{p \leq x : p\# + 1 \text{ prime}\} \sim e^\gamma \log x$.

Conjecture 11 ($n^2 + 2^n$; (b), sequence OEIS A064539). *All primes of the form $n^2 + 2^n$ with $n > 1$ have $n \equiv 3 \pmod{6}$, and their counting function satisfies*

$$\#\{n \leq N\} \sim \sum_{\substack{n \leq N \\ n \equiv 3(6)}} \frac{\kappa}{\log(n^2 + 2^n)}, \quad \kappa = \prod_p \frac{1 - \delta_p}{1 - 1/p} = 4.2734 \dots,$$

where δ_p is the density of $n \equiv 3(6)$ with $p \mid n^2 + 2^n$ in a full period $\text{lcm}(6, p \cdot \text{ord}_p 2)$. In particular infinitely many such primes exist, with $\sim (\kappa/6 \log 2) \log N$ of them up to N .

Conjecture 12 (Factorial primes; (a), [3]). $\#\{n \leq N : n! + 1 \text{ prime}\} \sim e^\gamma \log N$.

Verification. C10: 11 primes with $p \leq 4000$ against predicted 14.8 ($z = -0.98$). C11: hits $n \in \{3, 9, 15, 21, 33, 2007, 2127, 3759\}$ up to 4200; the mod-6 constraint was verified exhaustively (no prime off the lane to $n = 3000$, independently). C12: 15 primes with $n \leq 700$ against predicted 11.7 ($z = +0.98$). The paired statements C10 and C12 share a constant but not a sequence; the data straddle it ($z = -1.0$ and $z = +1.0$), which is what a correct shared model looks like.

4 Representation conjectures

Theorem 1 (Cube obstruction; found by adversarial audit). *For $k \geq 2$, the cube k^3 is representable as $p + j^3$ with $j \geq 1$ if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite, of counting function $\sim x^{1/3}$.*

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$ is composite unless $k - j = 1$, in which case it equals $3j^2 + 3j + 1 = 3k^2 - 3k + 1$ evaluated at $j = k - 1$. $\square$

Conjecture 13 (Prime plus a cube, restated; (a)-core, Hardy–Littlewood $E_3(X) = O(1)$ for the non-cube part). *(i) The set of non-cube integers $n \geq 2$ not representable as $p + k^3$ ($k \geq 1$) is finite; its per-decade counts decay super-geometrically, and the largest non-cube exception below 10^8 is 78,526,384. (ii) The representable cubes follow Bateman–Horn: $\#\{k \leq K : 3k^2 - 3k + 1 \text{ prime}\} \sim C(3x^2 - 3x + 1)I(K)$.*

Finding 1. As first stated (“the exception set of $n = p + k^3$ is finite”), this conjecture is **false**, and our own adversarial battery refuted it: all 412 exceptions found in $(10^8, 10^9]$ are perfect cubes, and Theorem 1 explains them. The error was a textbook violation of the algebraic-obstruction clause of the design criteria: cubes have density zero, so the Borel–Cantelli sum never noticed them. We record the failure rather than hide it; the restated version separates theorem from conjecture.

Conjecture 14 (Stern-type list; (a), OEIS A060003). *Exactly ten odd integers are not of the form $p + 2k^2$ with $k \geq 1$:*

$$1, 3, 17, 137, 227, 977, 1187, 1493, 5777, 5993.$$

(Verified here to 10^9 ; the seven prime members are the Stern primes.)

Conjecture 15 (Goldbach in the $(3, 3)$ lane; (a), K. Martin [7]). *Every $n \equiv 2 \pmod{4}$ with $n \geq 6$ is $p + q$ with $p \equiv q \equiv 3 \pmod{4}$ both prime, and the ordered count satisfies*

$$R_3(n) \sim \frac{1}{2} \mathfrak{S}(n) \int_3^{n-3} \frac{dt}{\log t \log(n-t)}, \quad \mathfrak{S}(n) = 2C_2 \prod_{\substack{p|n \\ p>2}} \frac{p-1}{p-2}.$$

(Parity forces $p \equiv q \equiv 1 \pmod{4}$ when $n \equiv 2 \pmod{4}$; the $(3, 3)$ lane is the one with no exceptions at all.)

Conjecture 16 (Uniform quantitative de Polignac; (a)-core). $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$ uniformly for even $d \leq (\log x)^A$; concretely, at any fixed large x the residuals z_d over even $d \leq 2000$ form a single mean-zero, unit-order Gaussian profile with regression slope 1 and no trend in d .

Conjecture 17 (Dubner’s conjecture; (a) [5]). *Every even $n \geq 4210$ is a sum of two members of twin-prime pairs; the 35 exceptions (OEIS A007534, largest 4208) are complete.*

Verification. C14, C15, C17: exhaustive to 10^8 , then adversarially re-swept over $(10^8, 10^9]$ with no new exception (C17 additionally re-verified to 10^9 by an independent sumset implementation). C15 representation counts at $n \approx 10^6, 10^7, 10^8$ match the formula to 0.2–1.3%. C16 at $x = 10^8$: over the 1000 values $d \leq 2000$, correlation 0.999999, slope 0.99966, $\max |z_d| = 1.80$; stressed to $d \leq 6000$ (3000 values), $\max |z_d| = 2.01$ — three thousand singular-series values spanning a factor 3, one regression line.

5 Statistical laws

Conjecture 18 (Twin-gap records; (a), Kourbatov [8]). *The maximal gap $G_t(x)$ between consecutive twin-prime starts up to x satisfies $G_t(x) \asymp \log^3 x$ with $\limsup_x G_t(x)/\log^3 x = 1/(2C_2) = 0.75735\dots$ to first order (Granville-type local corrections may lift the constant; the claim to take seriously is the cube of the logarithm).*

Conjecture 19 (First-occurrence gaps, Granville-corrected Shanks; (b)). *Let $p(g)$ be the prime starting the first occurrence of a prime gap of exactly g . Then*

$$\frac{\log p(g)}{\sqrt{g}} \longrightarrow \sqrt{e^\gamma/2} = 0.94358\dots,$$

approached from above. (Shanks' 1964 heuristic asserts limit 1; if maximal gaps obey Granville's $\limsup G/\log^2 p = 2e^{-\gamma}$, the first-occurrence law must follow the corrected constant instead. The two differ by 6%, below any feasible computation's resolution—the conjecture's value is that it registers a definite competing constant.)

Conjecture 20 (Microscopic variance law; (b); constant from Montgomery–Soundararajan [12]). *Fix $\lambda > 0$ and let X count primes in $[t, t + \lambda \log x)$ for t random near x . Then $X \Rightarrow \text{Poisson}(\lambda)$ (Gallagher [11]), and*

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + O\left(\frac{1}{\log^2 x}\right),$$

uniformly for fixed λ ; the additive constant is $\gamma + \log 2\pi - 1 = 1.41509\dots$, inherited from Montgomery's $\sum_{d \leq h} \mathfrak{S}(d)(h-d) = \frac{h^2}{2} - \frac{h}{2}(\log h + \gamma + \log 2\pi - 1) + o(h)$.

Remark 2. *The naive statement ($\text{Var}/\mathbb{E} = 1$) is refuted by the data at every accessible scale (deficits of 18–28%); the corrected law matches to three decimals with no fitted parameter: at $x = 10^9$, $\text{Var}/\mathbb{E} = 0.7910$ vs. predicted 0.7854 ($\lambda = 1$) and 0.7527 vs. 0.7520 ($\lambda = 2$); at $x = 4 \cdot 10^9$, 0.7998 vs. 0.7960 and 0.7670 vs. 0.7646. The residual +0.003 is the anticipated $O(1/\log^2 x)$ term.*

Conjecture 21 (Twin race modulo 5; (b), apparently new). *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod{5}$. Since $3 \mid q^2 + 2$ for every prime $q > 3$, the only twin patterns containing a prime square are $(q^2 - 2, q^2)$, and $q^2 - 2 \pmod{5} \in \{2, 4\}$, never 1. Consequently: (i) class 1 leads: with $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$,*

$$D_1(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2) + O^*(\sqrt{\pi_t(x)}),$$

the systematic term being governed by the constant C_7 of Conjecture 7; (ii) classes 2 and 4 are symmetric; (iii) since the bias is smaller than the noise by a factor $\log x$, the race has no persistent leader: unlike Chebyshev races, the logarithmic density of $\{D_1 > 0\}$ tends to a limit strictly between $\frac{1}{2}$ and 1.

Conjecture 22 (Least primes in progressions; (b); part (i) cf. [13]). *Let $p(a, q)$ be the least prime $\equiv a \pmod{q}$ and $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$. Then: (i) $U \Rightarrow \text{Exp}(1)$ as $q \rightarrow \infty$; in particular $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \leq n} 1/k$), the order-statistics form*

of Wagstaff’s $\max_a p(a, q) \asymp \varphi(q) \log^2 q$; (ii) the finite- q deficit is first-order $1/\log q$: $\mathbb{E}_a[U] = 1 - \theta(q)/\log q + o(1/\log q)$ with θ slowly varying (measured: $\theta \approx 1.5\text{--}1.9$ for $q \in [10^2, 6 \cdot 10^3]$, mean U bottoming near 0.737 around $q \approx 200$ and recovering through 0.776 by $q \approx 6000$).

Remark 3. *The deficit is not an artifact: a Cramér pseudo-prime control shows about half of it (discrete hazards), and the other half vanishes under any reshuffling of the actual prime residues—the ordering of the primes fills residue classes faster than any exchangeable model. The constant θ is measured, not derived; we consider it the deepest unexplained number produced by this exercise.*

Conjecture 23 (Fermat quotients; (a), cf. [15]). *With $q_p = (2^{p-1} - 1)/p \bmod p$: (i) q_p/p equidistributes on $[0, 1)$ with square-root discrepancy (KS distance $O(\pi(x)^{-1/2})$, no drift); (ii) $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$; (iii) hence $\#\{\text{Wieferich } p \leq x\} = \log \log x + O(1)$, so $\{1093, 3511\}$ below 10^8 and a third example only at doubly-exponential heights.*

Conjecture 24 (Conjecture F as a family; (a), Hardy–Littlewood [1], Fung–Williams, Jacobson–Williams [16]). *For odd A let $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then $Q_A(N) = C(A) I_A(N)(1 + o(1))$ uniformly over odd $A \leq 199$, with jointly Poisson-size residuals; in particular the computed constants predict the complete empirical ranking, and Euler’s $A = 41$ tops the family with $C(41) = 6.64092$.*

Conjecture 25 (Consecutive primes mod 3; (a), Lemke Oliver–Soundararajan [17]). *For consecutive primes $p, p' \leq x$ (both > 3), the same-residue fraction $s(x)$ satisfies*

$$\frac{1}{2} - s(x) = (c + o(1)) \frac{\log \log x}{\log x}, \quad c \in (0, \infty),$$

with the deficit split symmetrically between the (1, 1) and (2, 2) patterns. Measured: $\hat{c} = 0.400, 0.375, 0.370$ at $x = 10^7, 10^9, 4 \cdot 10^9$ (slow drift consistent with the $o(1)$; treat c as lying in $[0.30, 0.40]$ pending the secondary terms), and $\#(1, 1)/\#(2, 2) = 0.99989$ at $4 \cdot 10^9$.

Verification of the statistical block. C18: records to $4 \cdot 10^9$ (largest observed gap 5292 after the twin 2,466,641,069; normalized records wander in $[0.41, 0.56]$, approaching the asymptote from below on a $\log \log$ clock, as they must). C19: all first occurrences to $4 \cdot 10^9$; regression slope $1.307 \rightarrow 1.297$ from 10^9 to $4 \cdot 10^9$, falling as required, far from distinguishing 0.944 from 1—as predicted. C21 at 10^9 : class counts (1141217, 1142128, 1141159); D_1 stays within one noise unit of the systematic prediction, no persistent leader through $4 \cdot 10^9$, and the control difference $\pi_{t,2} - \pi_{t,4}$ wanders as a fair race. C22: pooled over all $q \leq 3000$ (2.7M classes), the upper tail of U is exponential and maxima track H_φ with Gumbel-size scatter (means 2.5 below H_φ at these q , the finite-size deficit of part (ii)). C23: KS at $x = 10^5, \dots, 10^8$: $\sqrt{n}$ KS = 0.82, 0.82, 0.52, 0.63 (bounded, no drift); small-quotient census at $K = 100$: 169 vs. 161.2; Wieferich = $\{1093, 3511\}$ at 10^8 against expected 2.67. C24: 100 polynomials at $N = 10^6$: mean ratio 1.0006, s.d. 0.0027, $\max |z| = 1.92$, rank correlation 0.99988; $A = 41$: observed 261,080 vs. predicted 261,071.9 ($z = +0.02$).

6 Adversarial audit

Beyond the primary verification, every falsifiable-by-instance statement was attacked by scripts hunting counterexamples a decade past the original bounds, and by clean-context agents recomputing constants and counts from the bare statements alone. Summary: C13 as first stated

was refuted (Finding 1) and is restated above; one memo claim about the history of C14 was falsified by the literature audit and corrected; everything else survived: no new exceptions for C14/C15/C17 to 10^9 ; uniformity for C16 to $d = 6000$; the C20 law within 0.4% at $4 \cdot 10^9$; the C22 recovery confirmed on $q \in (3000, 6000]$; C4 at $4 \cdot 10^9$ with $z = +0.05$. Independent recomputations agreed with every published constant within stated truncation error. The full audit trail (including the failures) ships with the repository.

7 Attribution

Fifteen of the twenty-five statements were found by the audit to be previously stated in essence: C1 (OEIS A080149), C3 [6], C6 (A188596), C9 [4], C10/C12 [3], C13-core (Hardy–Littlewood), C14 (A060003), C15 [7], C16-core (uniform Hardy–Littlewood B), C17 [5], C18 [8], C23 [15], C24 [1, 16], C25 [17]. For these, our contribution is the computed constants, the uniform machine verification, and the adversarial re-testing. The ten carrying instance-level novelty are C2, C4, C5, C7, C8, C11 (quantified forms and constants), and C19, C20, C21, C22 (statements we could not find anywhere); of these, C21 and C22(ii) are, to our knowledge, the most genuinely new claims in this paper.

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